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BRAC UNIVERSITY
Department of Computer Science and Engineering

CSE 260: Digital Logic Design
Md. Arifur Rahman [MAR]

Semester: Fall 2022
Full Marks: 15

Examination: Quiz 1
Duration: 30 Minutes

Answer the following questions. Show calculations where required.
Figures in the right margin indicate marks. Understanding the question is part of the examination.

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C01 Q1. Convert the following numbers to equivalent decimal numbers. 3

- a. $(1101001.1001)_2$
- b. $(2701)_8$

C01 Q2. Convert $(5412)_{10}$ to the following bases: 3

- a. Binary
- b. Hexadecimal

C01 Q3. What are ranges of decimal values we can represent in 8-bit binary using: 4

- a. Signed-magnitude
- b. 1's complement
- c. 2's complement
- d. Why can we represent an additional value using 2's complement compared to 1's complement?

C01 Q4. Perform $(1101)_2 - (1011)_2$ in: 5

- a. Regular binary representation
- b. 1's complement
- c. 2's complement